Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 10 · Part C: Electronics for Signal Processing Level 2

Track: Analog-to-Digital and Digital-to-Analog Conversion, Quantisation Noise, Dither, Oversampling, Sigma-Delta Modulation

Prerequisites: Problems 02, 07 · **Estimated effort:** 10–14 hours

ADC and DAC: Where the Analog World Becomes Numbers

Every bit costs money, power and telemetry bandwidth. Build a quantiser and measure $6.02B + 1.76$ dB, see why a quiet signal produces harmonics instead of noise and how dither cures it, extract ENOB and INL/DNL from simulated converters, trade speed for resolution with oversampling and a sigma-delta modulator, and choose a converter for a vibration channel under a 9.6 kbit/s downlink.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/10_adc_dac_and_quantisation/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii)

a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

An ADC performs two operations, sampling (problem 02) and quantisation: it replaces a continuous voltage by one of 2^B levels. The rounding error acts like additive noise, and the resulting signal-to-quantisation-noise ratio grows by about 6 dB per bit. Real converters add thermal noise, nonlinearity, clock jitter and missing codes, so the *effective* number of bits (ENOB) is always lower than the nominal B. A sigma-delta converter reaches 24 bits not by being extremely accurate but by sampling a one-bit quantiser very fast and shaping its noise out of the signal band.

The converter is also a data-rate decision. A spacecraft that digitises three accelerometer axes at 1 kHz with 16 bits produces 48 kbit/s, far above a UHF CubeSat link of 9.6 kbit/s; the converter choice and the onboard processing determine what can be downlinked at all.

Theoretical Framework

1. Uniform Quantisation

A mid-tread B-bit quantiser with full-scale range $[-V_{\text{ref}}, V_{\text{ref}}]$ has step $\Delta = 2V_{\text{ref}}/2^B$. When the signal is busy enough, the error is uniform on $[-\Delta/2, \Delta/2]$, white, uncorrelated with the signal, with power $\Delta^2/12$. For a full-scale sine ($P_s = V_{\text{ref}}^2/2$),

$$\text{SQNR} = \frac{V_{\text{ref}}^2/2}{\Delta^2/12} \;\rightarrow\; 6.02\,B + 1.76 \text{ dB}. \tag{1}$$

For other inputs the constant differs (a Gaussian input with loading factor V_{ref}/σ trades clipping against resolution).

2. Dither

For small or slowly varying signals the error is not noise-like; it is a distorted copy of the signal and appears as harmonics. Adding a small random *dither* before quantisation decorrelates the error. Triangular-PDF dither of ± 1 LSB has variance $\Delta^2/6$, so the total noise is $\Delta^2/12 + \Delta^2/6$ (4.77 dB more than undithered noise) but harmonics become a flat noise floor.

3. Dynamic Metrics, INL and DNL

SINAD is the ratio of signal power to everything else (noise and distortion), SFDR the ratio of the signal to the largest spur, and

$$\mathrm{ENOB} = \frac{\mathrm{SINAD} - 1.76}{6.02}. \tag{2}$$

Static errors: differential nonlinearity $\mathrm{DNL}[k] = (W_k - \Delta)/\Delta$ for code width W_k, and integral nonlinearity $\mathrm{INL}[k] = \sum_{j \le k} \mathrm{DNL}[j]$. DNL $= -1$ means a missing code. The *code-density* (histogram) test estimates the widths from the count of samples in each code when a slowly varying known input (a ramp or a sine) is applied.

4. Oversampling and Noise Shaping

Sampling at OSR× the Nyquist rate spreads the quantisation noise over a wider band; filtering to the signal band keeps only 1/OSR of it (+3.01 dB, i.e. half a bit, per doubling), provided the noise is white. A first-order sigma-delta modulator has noise transfer function $\mathrm{NTF}(z) = 1 - z^{-1}$; a second-order one $(1 - z^{-1})^2$. For an Lth-order modulator with an N-bit internal quantiser,

$$\mathrm{SQNR} \approx 6.02N + 1.76 - 10\log_{10}\frac{\pi^{2L}}{2L+1} + (20L + 10)\log_{10}\mathrm{OSR} \ \ \mathrm{dB}, \tag{3}$$

i.e. $9L + 3$ dB per octave of OSR.

5. DAC and Zero-Order Hold

A DAC holds each value for T_s, multiplying the spectrum by $\mathrm{sinc}(f/f_s)$: droop of -0.91 dB at $f_s/4$, -3.92 dB at $f_s/2$, and images around multiples of f_s that an analog reconstruction filter must remove.

Data and Simulation Specification

Quantiser tests	Full-scale sine $f = 1013.7$ Hz (choose a coherent frequency: an integer number of cycles, relatively prime to N, in $N = 2^{14}$ samples at $f_s = 48$ kHz), $B = 2, \ldots, 16$; Gaussian input with loading factors 2–5.
Dither	Sine at -40 dBFS, $B = 10$, undithered and TPDF dithered.
Non-ideal ADC	10 bits, input noise 0.5 LSB RMS, third-order INL bow of 1 LSB; 6-bit device with random DNL of 0.2 LSB RMS for the code-density test (ramp of 2^{20} samples).
Sigma-delta	First- and second-order, 1-bit, OSR $= 8, 16, 32, 64, 128$, input sine at $f_s/(4\,\mathrm{OSR})$ with amplitude 0.5.
DAC	1 kHz sine from $f_s = 8$ kHz and from 8× oversampled data.
Real data (optional)	Sound-card loop-back: play a 1 kHz sine from a tone generator into the line input (or record a tone from a speaker), capture at 16-bit and, if available, 24-bit and compute SINAD and ENOB with your own code.

Your Task

1. **Quantiser and SQNR (Stage 2).** Implement a B-bit mid-tread quantiser with clipping. Verify that the error is uniform with power $\Delta^2/12$ and plot SQNR against B for a full-scale sine against Eq. (1). Repeat for a Gaussian input and find the loading factor that maximises SQNR.
2. **Spectra, harmonics and dither.** Compute SINAD, SFDR and the spectrum of the quantised sine with coherent sampling. Show that at -40 dBFS and $B = 10$ the error is dominated by

harmonics, then add TPDF dither and verify the harmonics vanish and the noise floor rises by 4.77 dB (measure it).

3. **ENOB of a non-ideal converter.** Add thermal noise and INL bow to a 10-bit quantiser. Plot SINAD and ENOB against input amplitude from −60 to 0 dBFS and against input frequency if you add the jitter model of problem 02. Explain the ENOB-versus-amplitude curve.
4. **INL and DNL by code density.** Simulate the 6-bit converter with random code widths, apply the ramp, and recover DNL and INL from the histogram. Compare with the truth, identify any missing codes and quantify the estimator error versus number of samples.
5. **Oversampling.** Quantise a noisy sine at OSR $= 1, 4, 16, 64$ with 8 bits, low-pass filter and decimate. Verify the +3 dB per octave improvement when thermal noise exceeds 1 LSB and show that it fails without noise or dither.
6. **Sigma-delta modulator.** Implement the first- and second-order 1-bit modulators. Plot the bitstream spectrum showing noise shaping, decimate with an ideal brick-wall filter in the FFT domain (and later with a CIC filter in problem 16) and plot SQNR versus OSR. Fit the slopes (9 and 15 dB per octave) and compare with Eq. (3).
7. **DAC reconstruction.** Verify the ZOH droop at $f_s/4$ and $f_s/2$, display the images, and show how 8× oversampling relaxes the analog reconstruction-filter order. Design a short FIR that compensates the droop.

Self-check targets

$B = 12$: 74.0 dB; $B = 16$: 98.1 dB; $B = 24$: 146.2 dB (ideal). TPDF dither raises the noise floor by $10 \log_{10} 3 = 4.77$ dB. ZOH droop: −0.91 dB at $f_s/4$, −3.92 dB at $f_s/2$. For a *full-scale* sine at OSR = 64 Eq. (3) gives about 56.8 dB (first order) and 85.2 dB (second order). Your simulation will come out lower: driving the 1-bit loop at amplitude 0.5 costs 6 dB by itself, and the linear noise model is optimistic for a 1-bit quantiser, so values of roughly 45–51 dB (first order) and 68–79 dB (second order) are normal. The robust check is the *slope*: about 9 dB per octave of OSR for the first-order and 15 dB per octave for the second-order modulator.

Aerospace Application Link

An engine-test accelerometer channel must resolve 5 mg detail on top of ±50 g peaks, a dynamic range of about 80 dB in amplitude before any noise margin. Select the converter: compute the minimum ENOB and compare 16-bit SAR and 24-bit sigma-delta devices of your choice. Then compute the raw data rate of three axes at 1 kHz and the compression factor (decimation, event-triggered capture, feature extraction from problem 21) needed to fit a 9.6 kbit/s downlink, and the maximum record you can transmit during one 8-minute ground-station pass.

Deliverables

D1 **Quantiser notebook:** error statistics and the SQNR-versus-B plots for sine and Gaussian inputs.

D2 **Dither study:** spectra with and without dither, measured noise increase.

D3 **ENOB and INL/DNL:** the amplitude sweep and the code-density test with true versus estimated DNL/INL.

D4 **Oversampling and sigma-delta:** the 3-dB-per-octave verification, bitstream spectra and SQNR-versus-OSR fits.

D5 **DAC study:** droop, images and the compensation filter.

D6 **Memo (250 words):** "Which converter and which data-reduction scheme for the engine-test channel?" Cite at least two external sources.

Bonus (a) Implement a MASH 1-1 modulator and compare with the second-order loop. (b) Add a thermal-noise and jitter model and find the sampling frequency above which jitter dominates your ENOB. (c) Implement a successive-approximation ADC with capacitor mismatch and observe the resulting DNL pattern.

Learning Resources

Converters

- Kester, W. (ed.) *Data Conversion Handbook*, Analog Devices (quantisation, ENOB, dither, sigma-delta); free from the Analog Devices website.
- Schreier, R., Temes, G.C. *Understanding Delta-Sigma Data Converters* (noise shaping and SQNR formulas).
- IEEE Std 1241, *IEEE Standard for Terminology and Test Methods for Analog-to-Digital Converters*.

Theory and Practice

- Oppenheim, A.V., Schafer, R.W. *Discrete-Time Signal Processing*, Ch. 4.8–4.9 (quantisation noise, oversampling and noise shaping).
- Lipshitz, S.P., Wannamaker, R.A., Vanderkooy, J. (1992). "Quantization and dither: a theoretical survey." J. Audio Eng. Soc. 40, 355.

Tools

- NumPy FFT and SciPy signal: `https://numpy.org/doc/stable/reference/routines.fft.html`, `https://docs.scipy.org/doc/scipy/reference/signal.html`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.